UNIT V – ASSOCIATION RULE MINING & VISUALIZATION

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1. Concept of Association Rules

Definition

- Association Rule Mining discovers **relationships between items in large datasets**

Basic Form

```
X → Y
```

- X = Antecedent
- Y = Consequent

Example

- {Bread, Butter} → {Jam}

 Meaning: Customers who buy bread and butter also buy jam

Applications

- Market basket analysis
- Recommendation systems
- Web usage mining

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2. Frequent Itemset Mining – Apriori Algorithm

Concept

- Finds **frequent itemsets** based on minimum support

Apriori Principle

 “If an itemset is frequent, all its subsets must also be frequent”

Steps

1. Find frequent 1-itemsets
2. Generate candidate 2-itemsets
3. Prune infrequent sets
4. Repeat until no more itemsets

Advantages

- Simple
- Easy to understand

Limitations

- High computation
- Multiple database scans

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3. Support, Confidence, and Lift

Support

```
Support(X → Y) = Frequency(X ∪ Y) / Total Transactions
```

Confidence

```
Confidence(X → Y) = Frequency(X ∪ Y) / Frequency(X)
```

Lift

```
Lift = Confidence / Support(Y)
```

Interpretation

- Support → Popularity
- Confidence → Strength
- Lift → Dependency

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4. Large Itemsets and Candidate Generation

Concept

- Candidate itemsets are generated from frequent itemsets

Process

- Join step → combine itemsets
- Prune step → remove infrequent ones

Problem

- Large number of candidates → high cost

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5. Parallel and Distributed Mining

Need

- Large datasets require distributed processing

Techniques

- MapReduce
- Parallel Apriori

Advantages

- Faster processing
- Scalable

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6. Incremental Rule Mining

Concept

- Updates rules when new data is added

Advantages

- Avoids reprocessing entire dataset
- Efficient for dynamic databases

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7. Advanced Technique – FP-Growth

Concept

- Uses **FP-Tree (Frequent Pattern Tree)**

Steps

1. Build FP-tree
2. Extract frequent patterns

Advantages

- No candidate generation
- Faster than Apriori

Comparison

Feature	Apriori	FP-Growth
Candidate Generation	Yes	No
Speed	Slow	Fast

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8. Rule Quality Evaluation

Measures

- Support
- Confidence
- Lift
- Conviction

Goal

- Identify **useful and interesting rules**

Example

- High confidence but low support → less useful

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9. Visualization of Multidimensional Data

Purpose

- Represent complex data visually

Techniques

- Scatter plots
- Heat maps
- 3D visualization
- Parallel coordinates

Benefits

- Easy understanding
- Better decision making

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10. Tools – WEKA Case Study

WEKA Tool

- Open-source data mining tool

Steps in WEKA

1. Load dataset
2. Select **Associate tab**
3. Choose Apriori
4. Set support & confidence
5. Run algorithm

Output

- Association rules
- Rule strength

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11. Applications of Association Rule Mining

A. Market Basket Analysis

- Identify product combinations

B. Healthcare

- Disease pattern detection
- Drug interaction analysis

C. Finance

- Fraud detection
- Risk analysis

D. E-commerce

- Recommendation systems

□ Final Summary

- Association rule mining finds **hidden relationships**
- Key techniques:
    - Apriori

    - FP-Growth
- Metrics:
    - Support, Confidence, Lift
- Visualization helps in **interpretation**

□ Exam-Oriented Preparation

2-Mark Questions

- Define association rule
- What is support?
- What is Apriori?

16-Mark Questions

- Explain Apriori algorithm
- Compare Apriori and FP-Growth
- Describe rule evaluation metrics
- Explain WEKA association mining